Introduction to Data Analytics
MS - 4610

Session 11 – Classification and Regression Trees (CART)

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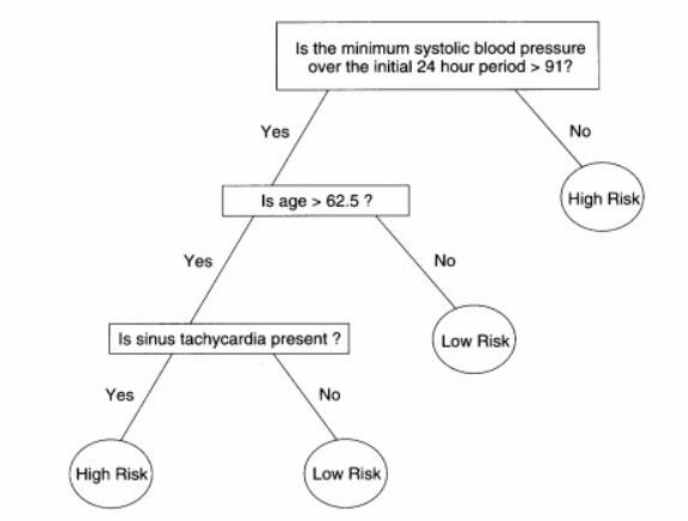
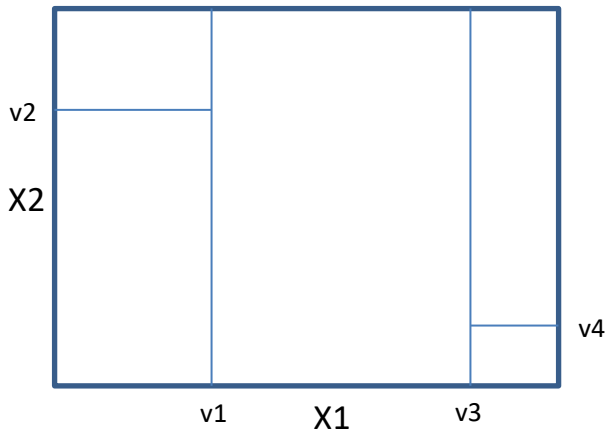
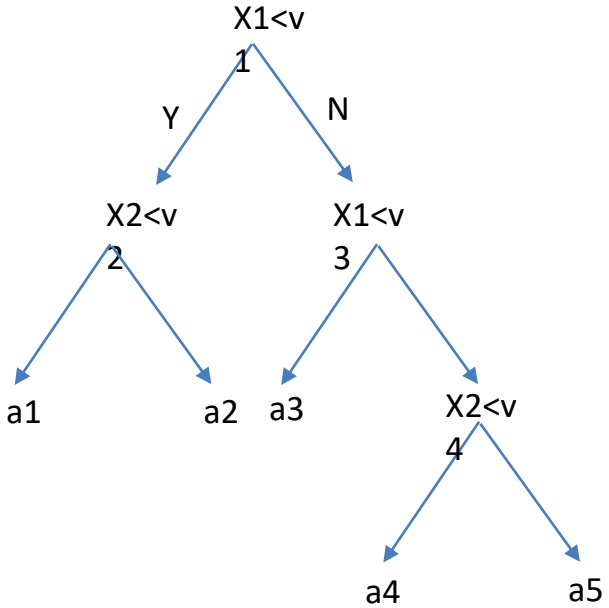
CART – How we look at heart-attacks differently

- The general linear model vs CART approach



CART- Classification and Regression Trees





The Math

- The minimization criterion for training the tree (the equivalent of OLS for Regression)
- Minimization criteria for a standard split using CART

$$\min_{j,s}\left[\min_{c_1}\sum_{x_i\in R_1(j,s)}(y_i-c_1)^2+\min_{c_2}\sum_{x_i\in R_2(j,s)}(y_i-c_2)^2\right]$$

- where j represents an input variable and s is the split point in the particular variable
- This defines the pair of half planes as

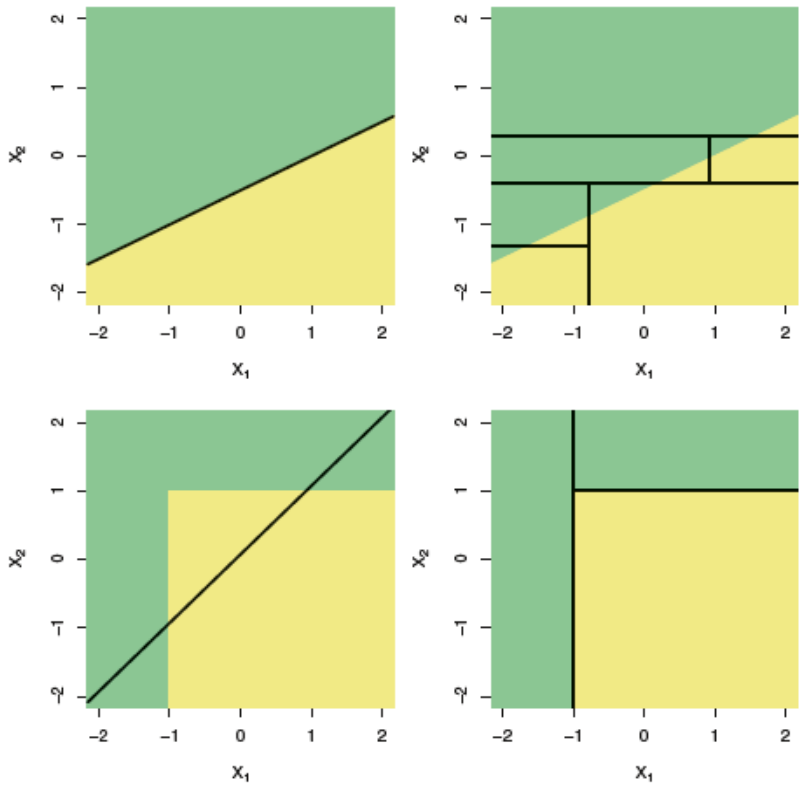
$$R_1(j,s)=\{X|X_j\leq s\} \qquad R_2(j,s)=\{X|X_j>s\}$$

- For any choice of j and s, the inner minimization solves to (for a regression problem)

$$\hat{C}_1=\text{ave}(y_i|X_i\in R_1(j,s)) \qquad \hat{C}_2=\text{ave}(y_i|X_i\in R_2(j,s))$$

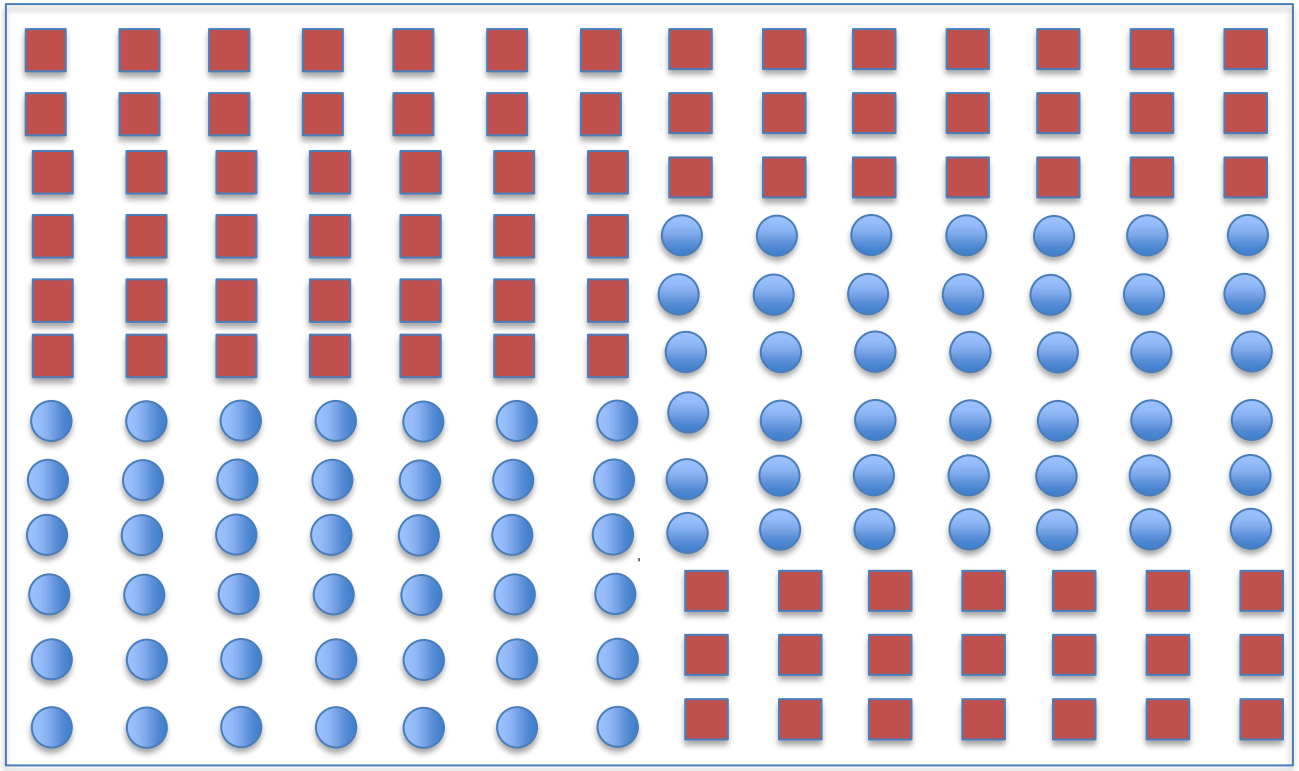
Trees

- Where does it work:



Trees

- Where does it work, where does it fail?



Overfitting and Underfitting trees

- Minimum leafs per node
- Cost-complexity Pruning with validation or cross validation (number of terminal nodes)

$$\sum_{m=1}^{|T|} \sum_{i:\ x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$$

- Do this for all $T \subset T_0$, where $T_0$is the large tree
- $|T|$indicates the terminal nodes of a tree T
- Why is pruning better than not letting the tree grow?

Advantages and Disadvantages

- From ISLR:

▲ Trees are very easy to explain to people. In fact, they are even easier to explain than linear regression!

▲ Some people believe that decision trees more closely mirror human decision-making than do the regression and classification approaches seen in previous chapters.

▲ Trees can be displayed graphically, and are easily interpreted even by a non-expert (especially if they are small).

▲ Trees can easily handle qualitative predictors without the need to create dummy variables.

▼ Unfortunately, trees generally do not have the same level of predictive accuracy as some of the other regression and classification approaches seen in this book.

- The nature of greedy algorithms. Why do we need random forests, bagging, boosting.